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Abstract

This project aims to develop a structured and curated peptide database by resolving unknown amino acids and expanding incomplete functional annotations in UniProt peptide sequences. The study will combine external peptide databases, literature verification, and UniRef cluster information to improve sequence quality and annotation reliability. Machine learning models will then be used to study how ambiguous sequence information affects peptide function prediction. The project will also develop uncertainty-aware prediction methods to improve prediction reliability under incomplete data conditions. The expected outcome is a high quality peptide resource that supports more accurate and reliable AI-based peptide discovery and function prediction.

Project Narrative

Protein and peptide databases are important resources for biomedical research and drug discovery. However, many peptide sequences contain unknown amino acids and incomplete functional annotations, which reduce the reliability of artificial intelligence models. This project will create a curated peptide database with improved sequence quality and expanded annotations to support reliable peptide function prediction. The proposed work will improve computational peptide analysis and may support future development of antimicrobial and therapeutic peptides.

References

1. Consortium, U. UniProt: a worldwide hub of protein knowledge. *Nucleic acids research* **47**, D506–D515 (2019).
2. Pirtskhalava, M. *et al.* DBAASP v3: database of antimicrobial/cytotoxic activity and structure of peptides as a resource for development of new therapeutics. *Nucleic acids research* **49**, D288–D297 (2021).
3. Wang, G., Li, X. & Wang, Z. APD3: the antimicrobial peptide database as a tool for research and education. *Nucleic acids research* **44**, D1087–D1093 (2016).
4. Shi, G. *et al.* DRAMP 3.0: an enhanced comprehensive data repository of antimicrobial peptides. *Nucleic acids research* **50**, D488–D496 (2022).

Facilities and Equipment

During this proposed work, I will have access to office space and computing facilities at the University of Central Florida (UCF). The project will primarily utilize the UCF Advanced Research Computing Center (ARCC) high-performance computing cluster for large-scale data processing, machine learning model training, and storage. In addition, institutional storage servers will be used for secure data management and backup. A dedicated local workstation (Dell Precision 7865 Tower) with high memory and GPU capability will be used for data preprocessing, feature computation, and initial model development before scaling to ARCC. These resources provide sufficient infrastructure to complete the proposed research efficiently without requiring major new equipment purchases.

Biosketch

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.

Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Tapotee, Malisha Islam

eRA COMMONS USER NAME (credential, e.g., agency login): ma689713

POSITION TITLE: Graduate Student Research Assistant

EDUCATION/TRAINING (Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)

INSTITUTION AND LOCATION	DEGREE (if applicable)	Start Date MM/YYYY	Completion Date MM/YYYY	FIELD OF STUDY
University of Dhaka, Bangladesh	BSc	01/2018	02/2023	Electrical Engineering
University of Central Florida	PhD	01/2024	01/2028 (Expected)	Electrical Engineering

A. Personal Statement

I first became interested in research during my undergraduate studies in Electrical and Electronic Engineering at the University of Dhaka, where I was introduced to machine learning and its applications in healthcare. I worked under the supervision of Dr. Md Atiqur Rahman Ahad on human activity recognition and participated in international competitions, which strengthened my skills in data analysis and model development. Later, I collaborated with Dr. Muhammad E. H. Chowdhury on a senior-year project focused on physiological signal analysis using deep learning for wearable health monitoring. These experiences sparked my interest in applying artificial intelligence to biomedical problems. Currently, my predoctoral research focuses on generating antibacterial peptides using large language models and narrowing down promising candidates through a deep learning-based screening system. Thirteen peptides were selected for experimental validation against Gram-positive and Gram-negative bacteria, and three have already shown promising results. Through this work, I am gaining strong interdisciplinary training in computational biology and machine learning. I hope to submit this work as a first-author publication in the coming months, and I am committed to building a long-term career in research focused on AI-driven biomedical discovery.

B. Positions, Scientific Appointments and Honors

Positions and Scientific Appointments

2024 – Present	Graduate Research Assistant, University of Central Florida
2021	Engineering Intern, Datasoft Manufacturing & Assembly Inc. Limited, Bangladesh
2018 – 2022	Member, IEEE Student Branch University of Dhaka
2018	Volunteer, FabLab, University of Dhaka, Bangladesh

Honors

2025	Poster Presenter, Biomedical Engineering Society 2025 Annual Meeting
2022	Virtual Outstanding Participation, IAPR/IEEE Winter School on Biometrics 2022
2021	2nd Best Paper, IEEE Computer Society Bangladesh Chapter Winter Symposium 2021
2021	2nd Runner-up, 3rd International Conference on Activity and Behavior Computing 2021
2021	1st Runner-up, Sussex-Huawei Locomotion and Transportation Recognition Challenge 2021
2021	2nd Runner-up, Ada Lovelace Datathon 2021

C. Contributions to Science

1. Undergraduate Research

During my undergraduate studies in Electrical and Electronic Engineering at the University of Dhaka, I gradually developed a strong interest in machine learning and its real-world applications. I worked under the supervision of Dr. Md Atiqur Rahman Ahad, where I contributed to research on human activity recognition and participated in international competitions, gaining hands-on experience in data preprocessing, feature extraction, and model development. I also collaborated with Dr. Muhammad E. H. Chowdhury on a senior-year project involving physiological signal analysis, where we used deep learning models to transform wrist photoplethysmogram signals into finger photoplethysmogram signals for wearable health monitoring. These research experiences strengthened my technical skills and sparked my long-term interest in applying artificial intelligence to healthcare and biomedical problems.

1. Tapotee, M. I., Saha, P., Mahmud, S., Alqahtani, A., & Chowdhury, M. E. (2024). M2ECG: Wearable mechanocardiograms to electrocardiogram estimation using deep learning. *IEEE Access*, 12, 12963–12975.
2. Saha, P., Tapotee, M. I., & Ahad, M. A. R. (2021, November). Task detection of ASD children by analyzing robotic enhanced and standard human therapy. In *2021 Thirteenth International Conference on Mobile Computing and Ubiquitous Network (ICMU)* (pp. 1–6). IEEE.
3. Basak, P., Tasin, S. M., Tapotee, M. I., Sheikh, M. M., Sakib, A. N., Baray, S. B., & Ahad, M. A. R. (2020, September). Complex nurse care activity recognition using statistical features. In *Adjunct Proceedings of the 2020 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2020 ACM International Symposium on Wearable Computers* (pp. 384–389).

2. Graduate Research

My ongoing predoctoral research focuses on antibacterial peptides. The primary objective of this work is to generate novel antibacterial peptide sequences using large language models. These generated candidates are then filtered and prioritized through a deep learning–based screening system to identify the most promising sequences. From this process, 13 peptide sequences were selected for experimental validation against both Gram-positive and Gram-negative bacteria. The most promising candidates are subsequently evaluated for cell viability to assess safety and therapeutic potential. To date, three peptides have been experimentally tested, yielding encouraging results. I anticipate submitting this work as a first-author publication in the coming months.

D. Scholastic Performance

University of Dhaka

YEAR	COURSE TITLE	GRADE
2018	Electrical Circuit Analysis I	A
2018	Computer Programming	A+
2018	Electricity and Magnetism	A+
2018	Differential and Integral Calculus	A+
2018	Vector Analysis	A+
2018	Computer Programming Laboratory	A
2018	Chemistry	B+
2018	Linear Algebra	B-
2018	Electrical Circuit Laboratory	A+
2018	Electric Circuit Simulation Laboratory	A+
2019	Physical Electronics and Semiconductor	A+
2019	Analog Electronics I	A
2019	Analog Electronics II	B+
2019	Analog Electronics Laboratory	A
2019	Digital Electronics	A+
2019	Digital Electronics Laboratory	B
2019	English for Technical Communication	A
2019	Statistics and Probability	A
2019	Fundamentals of Economics	A+
2019	Fundamentals of Mechanical Engineering	A+
2019	Differential Equations	A+
2019	Mechanical Engineering Laboratory	A+
2021	Electromagnetic Theory	A+
2021	Signals and Systems	A
2021	Electronic Devices	A
2021	Numerical Technique Laboratory	A+

2021	Microprocessor and Interfacing Laboratory	A+
2021	Communication Laboratory	A+
2021	Power System I	A+
2021	Power System I Laboratory	A+
2021	Digital Signal Processing	B
2021	Digital Signal Processing Laboratory	A+
2021	Electrical Services Design and Drafting	A+
2021	Financial Accounting and Cost Management	A
2021	Industrial Management	A+
2022	Material Science	A
2022	Database System	A+
2022	Intelligent System	A
2022	Computer Networks	A
2022	Computer Network Lab.	A+
2022	Professional Ethics	A
2022	Microwave and Satellite Communication	A
2022	Senior Project I	A
2022	Senior Project II	B+

University of Central Florida

YEAR	COURSE TITLE	GRADE
2024	Machine Learning	A
2024	Lab Virtual Simulations Physiology	A
2024	Fabrication of Solid-State Devices	B
2024	Biomaterials	A
2024	Introduction to Nanoscience and Nanotechnology	A
2025	Introduction to Neural Networks and Deep Learnings	A-
2025	Biomedical Nanotechnology	A
2025	Nanofabrication and Characterization	A
2025	MEMS Fabrication Laboratory	A
2025	Properties of Nano Materials	A

Budget

4. BUDGET JUSTIFICATIONS and Sample Calculations for the Initial Budget Period:

Sample completed PHS 398 Form for NIH Grants:

Principal Investigator/Program Director (Last, first, middle): Dr. Kausik Mukhopadhyay

DETAILED BUDGET FOR INITIAL BUDGET PERIOD DIRECT COSTS ONLY	FROM 07/01/26	THROUGH 06/30/26
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PERSONNEL (Applicant organization only)

NAME	ROLE ON PROJECT	TYPE APPT. (months)	% EFFORT ON PROJ.	INST. BASE SALARY	DOLLAR AMOUNT REQUESTED (omit cents)		
					SALARY REQUESTED	FRINGE BENEFITS	TOTALS
Dr. Kausik Mukhopadhyay	Principal Investigator	9	15%	\$123,456	\$18,518	\$3,537	\$22,055
Dr. Kausik Mukhopadhyay	PI	3	30%	\$41,152	\$12,346	\$2,358	\$14,704
Person X	Postdoctoral Fellow	12	40%	\$65,315	\$26,126	\$3,919	\$30,045
Malisha Islam Tapotee	Res. Asst.	12	100%	\$30,000	\$30,000	\$1,500	\$31,500
Person Y	Res. Asst.	12	100%	\$30,000	\$30,000	\$1,500	\$31,500
Person Z	Software Engineer	12	100%	\$70,000	\$70,000	\$10,500	\$80,500
Subtotal from x-personnel sheet					0	0	0
SUBTOTALS						\$186,990	\$23,314
\$210,304							

CONSULTANT COSTS Bioinformatics Consultant (5 days @ \$250 per day; also \$1,500 for travel, accommodations, and per diem) See Detailed Budget Justification	\$2,750
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EQUIPMENT (Itemize) Dell Precision 7865 Tower AMD Threadripper PRO 5975WX 256GB 2TB M.2 RTX A4000 See Detailed Budget Justification	\$6,500
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SUPPLIES (Itemize by category) Secure Cloud Storage & Backup: \$4,000 Office and Printing Supplies: \$1,500 API Access & Database Subscription Fees: \$3,000 Data Archival & DOI Registration Fees: \$2,000 Software Licenses (Bioinformatics tools): \$2,000 See Detailed Budget Justification	\$12,500
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TRAVEL Local in-state travel: \$2,000 Out-of-state travel: \$5,500 See Detailed Budget Justification.	\$7,500
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PATIENT CARE COSTS	
INPATIENT	0
OUTPATIENT	0

ALTERATIONS AND RENOVATIONS (Itemize by category)	0
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OTHER EXPENSES (Itemize by category) Open Access Publication Fees: \$4,000 Website Maintenance & Security: \$1,500 Web Hosting and Database Deployment: \$2,500 See Detailed Budget Justification	\$8,000
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SUBTOTAL DIRECT COSTS FOR INITIAL BUDGET PERIOD	\$ 247,554
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CONSORTIUM/CONTRACTUAL COSTS	
	DIRECT COSTS
\$15,000	INDIRECT COSTS
\$3,750	

TOTAL DIRECT COSTS FOR INITIAL BUDGET PERIOD (Item 7a, Face Page)	\$ 266,304
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PHS 398 (Rev.05/01)

Page _____ Form Page 4

a. PERSONNEL

i. Salary for faculty member with 9-month academic appointment:

PERSONNEL (Applicant organization only)					DOLLAR AMOUNT REQUESTED (Omit cents)		
		Type Appt	% Effort on Project	Institutional Base Salary	Salary Requested	Fringe Benefits	TOTALS
Dr. Kausik Mukhopadhyay	PI	9	15%	\$123,456	\$18,518	\$3,537	\$22,055
Dr. Kausik Mukhopadhyay	PI	3	30%	\$41,152	\$12,346	\$2,358	\$14,704

• **9-mo Salary Requested:**

▸ $\$123,456 \times 15\% = 18,518$

• **Summer Salary Requested:**

▸ $[(\$123,456 / 9 \text{ mo}) \times 3 \text{ mo}] \times 30\% = [41,152] \times 30\% = 12,346$

ii. Fringe Benefits for faculty member with 9-month academic appointment:

• 9-mo Academic Year: $(A+B+C+D) \times \text{Salary}$

$19.10\% \times 18,518 = 3,537$

- Summer: $(A+B+C+D) \times \text{Salary} = (19.10\%) \times 12,346 = 2,358$

Justification:

Dr. Kausik Mukhopadhyay – Principal Investigator. Dr. Mukhopadhyay will provide overall scientific leadership for the project, oversee UniProt data extraction and ambiguous amino acid resolution strategies, supervise machine learning model development, and guide database architecture and dissemination. Effort is 15% during the academic year and 30% during summer months, consistent with institutional policies.

b. CONSULTANT COSTS

CONSULTANT COSTS	2,750
Bioinformatics Consultant (5 days @ \$250 per day; also \$1,500 for travel, accommodations, and per diem)	

Sample Calculation: $[5 \text{ days} \times \$250/\text{day} + \$1,500] = 2,750$

Justification:

Consultants – A bioinformatics consultant will assist with resolving ambiguous amino acids (X residues) and validating annotation mapping methods.

c. EQUIPMENT

EQUIPMENT	6,500
Dell Precision 7865 Tower AMD Threadripper PRO 5975WX 256GB 2TB M.2 RTX A4000	

Justification:

EQUIPMENT – The budget includes a Dell Precision 7865 Tower workstation for local computational development. The estimated cost is \$6,500. This system will include 128 GB RAM, multi-core processor, large SSD storage, and a GPU. It will be used for preprocessing UniProt data, feature computation, and machine learning prototyping before scaling to UCF's ARCC cluster. Since ARCC is used for large-scale training, only one modest workstation is requested.

d. SUPPLIES

SUPPLIES	12,500
Secure Cloud Storage & Backup: \$4,000; Office and Printing Supplies: \$1,500; API Access & Database Subscription Fees: \$3,000; Data Archival & DOI Registration Fees: \$2,000; Software Licenses (Bioinformatics tools): \$2,000 See detailed budget justification.	

Justification:

Supplies include computational and operational costs.

Secure cloud storage and backup are budgeted at \$4,000 to maintain secure copies of UniProt datasets and derived feature tables.

API access and database subscription fees are budgeted at \$3,000 to allow automated data retrieval and cross-database validation.

Data archival and DOI registration fees are budgeted at \$2,000 to ensure the curated peptide database is publicly archived.

Software licenses for bioinformatics tools are budgeted at \$2,000.

Office and printing supplies are budgeted at \$1,500.

e. TRAVEL

TRAVEL Local in-state travel: \$2,000; Out-of-state travel: \$5,500. See detailed budget justification.	7,500
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Justification:

Local in-state travel is budgeted at \$2,000 for collaborative meetings and institutional coordination.

Out-of-state travel is budgeted at \$5,500 for conference dissemination. This includes airfare (\$800), hotel (\$900), registration (\$900), and per diem (\$300) per person, multiplied appropriately for key project personnel.

f. OTHER EXPENSES

OTHER EXPENSES Bulk printing: \$2,721; General Other: \$3,500; School Incentives: \$12,000; Participant Costs: \$1,100	\$19,321
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Justification:

Open-access publication fees are budgeted at \$4,000 to ensure public access to results.

Web hosting and database deployment are budgeted at \$2,500.

Website maintenance and security are budgeted at \$1,500.

g. CONSORTIUM/SUBCONTRACT COSTS

NIH PHS 398 Example:

Consortium/	Direct Costs	\$ 15,000
Subcontract Costs	Facilities and Administrative Costs*	\$ 3,750

*The approved F&A cost rate at subcontractor institution is only 25% in this example. The University's (i.e., UCF) F&A rate will be applied to both the consortium/subcontractor Direct Costs and F&A Costs.

Justification

An external collaborating institution will provide statistical validation support. The consortium includes \$15,000 in direct costs, with indirect costs calculated at 25% ($\$15,000 \times 0.25 = \$3,750$), resulting in a total consortium cost of \$18,750.

Specific Aims

Protein sequence databases are essential for biological and biomedical research. **UniProt** is one of the most widely used resources for protein and peptide data. It provides sequences, limited functional annotations, and references to literature. **However, many peptide sequences contain unknown amino acids marked as X.** The identity of these amino acids is often not specified. Functional annotations are frequently incomplete or listed as a single activity. Supporting literature sources are not always clearly linked. These issues reduce data reliability. They limit the use of UniProt in machine learning and deep learning studies. **There is a critical need for a structured and enriched peptide database with resolved amino acids and comprehensive functional annotations.**

Our long term goal of this work is to enable reliable peptide function prediction using artificial intelligence. The objective of this proposal is to resolve ambiguous amino acids and incomplete functional annotations in peptide sequences. **Proteins are composed of 20 standard amino acids. In UniProt, non standard or modified amino acids are frequently represented as X.** This obscures the true sequence composition. Functional annotations are often incomplete or limited to a single activity. Links to supporting literature are inconsistently provided. The rationale for this project is that data ambiguity reduces the accuracy of computational models. **The central hypothesis is that resolving unknown amino acids and expanding functional labels will improve prediction performance.** The expected outcome is a curated peptide resource optimized for machine learning. The payoff is improved peptide discovery and functional screening.

Specific Aim 1: Resolve unknown amino acids and expand functional annotations

Peptide sequences with unknown amino acids can be resolved using external resources and learned patterns.

We will implement Python workflows to extract peptide level data from UniProt. Sequences containing X will be identified and separated. We will cross check sequences with external databases and literature. Resolved amino acids will be added when evidence exists. Multiple functional activities will be recorded as separate fields. Each annotation will include a reference source. Sequences will retain their UniRef cluster identifiers. We will design the database to support queries by cluster membership. Functional annotations will be linked to each cluster. This will enable access to peptide sequences by both cluster and activity. We will develop a search engine for peptide sequence input.

A structured table with resolved amino acids and multi activity annotations.

Specific Aim 2: Quantify and model the impact of missing and ambiguous peptide information on function prediction

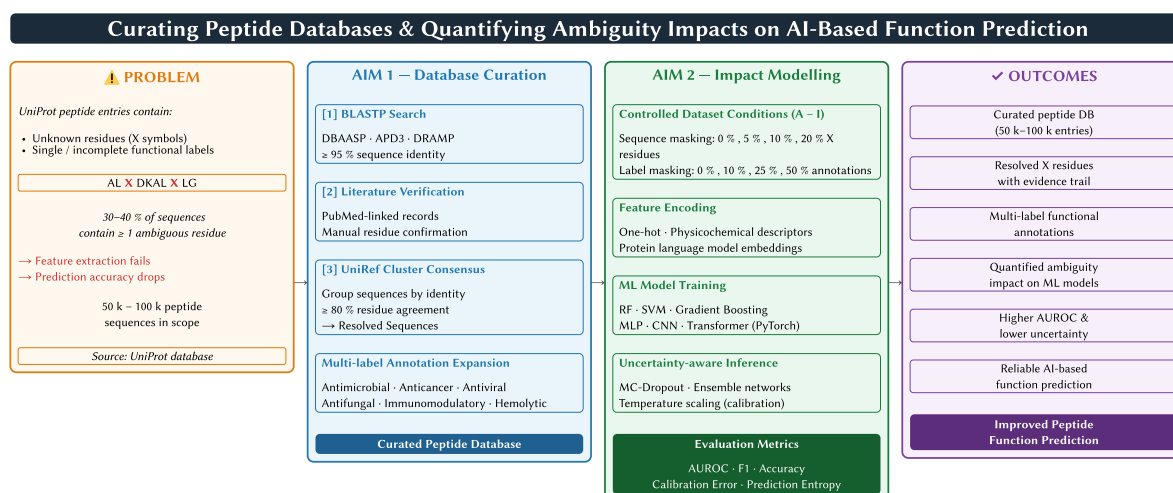
In this aim, we will study how incomplete sequence information and missing functional annotations affect peptide function prediction.

We will create datasets with different levels of ambiguity, including sequences with unknown amino acids (X), incomplete annotations, and missing metadata. We will train and evaluate machine learning models on these datasets to measure how ambiguity affects prediction accuracy and confidence. We will also develop methods to handle uncertainty, such as ambiguity-aware feature encoding and confidence-based scoring. Model performance will be compared across datasets with different levels of ambiguity. Improvement will be measured by improved stability, calibration, and accuracy under ambiguous conditions.

This aim will provide a clear understanding of how missing peptide information affects AI predictions and will produce a computational framework that improves prediction reliability when data are incomplete.

This project will produce a curated and structured peptide database. It will improve data usability for AI models. The approach integrates data curation with machine learning. The resource will reduce the need to search multiple databases. The database will be extensible as new data become available.

Research Strategy



Data sources: UniProt · DBAASP · APD3 · DRAMP · UniRef | Encodings: one-hot, physicochemical, transformer embeddings | Validation: bootstrap CIs, 5-fold CV, paired t-tests

Figure 1: Graphical Abstract of Proposed Aims.

Approach

Preliminary Data

We developed a reproducible analysis pipeline using reviewed UniProt protein sequences that contain unknown amino acid residues (X). The pipeline automatically builds a SQLite database, extracts sequence features, performs statistical analysis, and generates figures through a containerized workflow. Figure 2(a) shows the relationship between protein sequence length and the number of unknown residues. Most proteins contain only a small number of unknown residues, while a few proteins contain much larger uncertainty levels. This plot helps identify how sequence ambiguity changes across proteins of different lengths. Figure 2(b) shows a UMAP projection of the protein sequences based on sequence-level similarity. The colors represent proteins with single or multiple unknown residues, and point size represents the number of unknown residues. This visualization helps explore whether proteins with higher uncertainty form similar sequence-level patterns or clusters in the dataset.

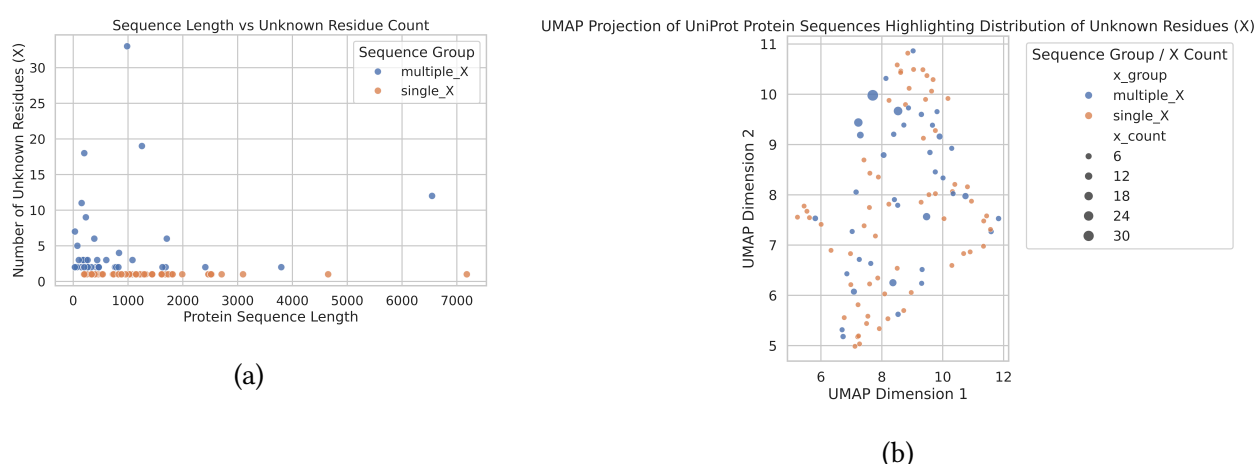


Figure 2: (a) Protein sequence length versus unknown residue count (X) in UniProt proteins. (b) UMAP projection of UniProt protein sequences containing unknown residues (X).

Specific Aim 1: Resolve unknown amino acids and expand functional annotations

Rationale: Peptide sequence databases are widely used in biological and biomedical research. Among them, UniProt¹ is one of the most comprehensive resources for protein and peptide data. It provides

sequence information, functional annotations, and references to supporting literature. However, many peptide entries contain incomplete information. In particular, some peptide sequences include unknown amino acids represented by the **symbol X**, which indicates that the identity of the residue is not known. In addition, functional annotations are often incomplete and frequently limited to a single reported activity, even though many peptides exhibit multiple biological functions.

These issues reduce the reliability of peptide databases and limit their usefulness for computational studies. Machine learning models depend on accurate sequence information and reliable labels. When peptide sequences contain unknown residues or incomplete annotations, feature extraction becomes unreliable and prediction accuracy decreases. As a result, ambiguous data can negatively affect computational models that attempt to predict peptide functions.

The objective of this aim is to create a structured and curated peptide dataset by resolving unknown amino acids and expanding functional annotations.

Overall Study Design: This study will develop a structured peptide database by resolving unknown amino acids and expanding incomplete functional annotations in UniProt peptide sequences using external databases, literature verification, and UniRef cluster information. The curated dataset will then be used to create multiple versions of peptide datasets with different levels of sequence ambiguity and annotation completeness. The results will help establish a reliable computational framework for peptide function prediction using high-quality sequence data.

1.1 Data Extraction and Preprocessing. Peptide sequence data will be extracted from the UniProt database using automated Python workflows. Data will be retrieved through the UniProt REST API and bulk downloads. The extraction pipeline will be implemented using the Python programming language with libraries such as Biopython, pandas, and requests. Each peptide entry will include information such as the UniProt accession identifier, amino acid sequence, sequence length, functional annotations, literature references, and UniRef cluster identifiers. The extracted dataset will be stored in a structured relational format to allow efficient querying and data management. Table 1 summarizes the key data fields that will be stored in the database.

Field	Description
UniProt ID	Unique peptide identifier
Sequence	Amino acid sequence
Length	Number of residues
Functional annotations	Biological activities
Literature references	Supporting publications
UniRef cluster ID	Sequence similarity cluster
Ambiguity flag	Presence of unknown residues

Table 1: UniProt Database Key data fields Description.

1.2 Identification of Ambiguous Peptide Sequences. After the initial dataset is extracted, peptide sequences will be analyzed to identify unknown amino acids. In UniProt records, unknown residues are represented by the symbol X. Each peptide sequence will therefore be scanned to detect the presence and positions of these ambiguous residues. Sequences will be separated into two groups: those with fully defined amino acids and those containing one or more unknown residues. For ambiguous sequences, the positions of the unknown residues will be recorded in the database. Preliminary analysis of peptide databases suggests that approximately 30–40% of peptide sequences contain at least one ambiguous residue. Identifying these sequences is an important first step because unknown

residues interfere with feature extraction and downstream modeling. Once identified, these ambiguous sequences will undergo a resolution procedure described below.

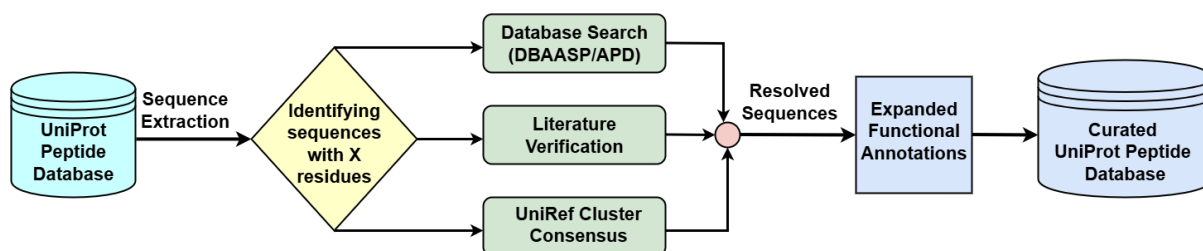


Figure 3: **Workflow for peptide sequence curation and annotation expansion.**

1.3 Resolution of Unknown Amino acids. Unknown amino acids will be resolved using a combination of sequence comparison, literature verification, and cluster-based inference. **The overall workflow of this process is illustrated in Figure 3.** First, each ambiguous peptide sequence will be compared with entries in external peptide databases. Databases such as DBAASP², APD³, and DRAMP⁴ contain curated antimicrobial and bioactive peptide sequences. Sequence similarity searches will be performed using BLASTP. If a sequence match is found with high similarity and the corresponding residue is known, the unknown amino acid will be replaced. Sequence identity thresholds of approximately 95% or greater will be used to ensure reliable residue replacement.

Second, peptide sequences with associated publications will be verified through literature searches. UniProt records often include PubMed references that describe the peptide sequence. When a sequence is reported in the literature with a clearly defined amino acid at the ambiguous position, the residue will be updated accordingly. All such updates will include a reference to the supporting publication.

Third, UniRef cluster information will be used to infer residues when direct evidence is not available. UniRef clusters group similar sequences based on sequence identity. If a peptide belongs to a cluster in which most sequences share the same residue at the ambiguous position, the consensus amino acid can be inferred. Residue replacement will be performed when the cluster consensus exceeds approximately 80% agreement.

This multi-step resolution strategy allows ambiguous residues to be resolved using complementary sources of evidence.

1.4 Functional Annotation Expansion. Functional annotations in UniProt often include only a single reported activity, even though many peptides exhibit multiple biological functions. To address this limitation, functional labels will be expanded by integrating information from multiple databases and literature sources. Each peptide will be assigned binary labels for different functional activities. Examples include antimicrobial, anticancer, antiviral, antifungal, immunomodulatory, hemolytic, and toxic activities. These annotations will be recorded as independent fields in the database so that peptides can be associated with multiple biological functions simultaneously. Functional labels will be collected from UniProt records, external peptide databases, and literature reports. When a functional activity is assigned to a peptide, the supporting reference will be stored in the database to ensure traceability. This approach allows the database to capture the multi-functional nature of many peptides.

1.5 Database Structure and Query System. A structured relational database will be constructed to organize the curated peptide data. The database will include several interconnected tables, including peptide sequences, functional annotations, literature references, and UniRef cluster information. This design will allow users to query the database using multiple criteria. For example, users will be able to search for peptides by sequence similarity, biological activity, or cluster membership. A simple search interface will be developed to allow researchers to input a peptide sequence and retrieve similar sequences and associated annotations. The database will also support clustering-based queries.

Because UniRef cluster identifiers will be retained for each peptide, researchers will be able to retrieve peptides belonging to the same cluster and examine their functional properties.

1.6 Sample Size and Statistical Considerations. The curated dataset is expected to include approximately 50,000 to 100,000 peptide sequences collected from UniProt and related resources. The success of the residue resolution pipeline will be evaluated by measuring the proportion of ambiguous residues that can be confidently resolved. Agreement between different resolution methods will also be assessed. Bootstrap sampling methods will be used to estimate confidence intervals for resolution accuracy and annotation consistency.

Controls and Validation: Positive control sequences with well-defined residues reported in literature will be used to verify that the pipeline correctly identifies and resolves ambiguous residues. In addition, negative control sequences containing artificially inserted unknown residues will be used to test whether the algorithm incorrectly assigns amino acids. Cross-database comparisons will also be performed to verify that resolved sequences match entries in external peptide databases. These validation steps will ensure that the curated dataset maintains high data integrity.

Expected Outcomes: The expected outcome of this aim is a curated peptide dataset with improved sequence accuracy and expanded functional annotations. Some peptide sequences may remain unresolved if sufficient evidence is not available. In such cases, ambiguous residues will be retained but marked as uncertain. This information will be preserved so that machine learning models can incorporate uncertainty during training. If literature evidence is insufficient to resolve certain residues, additional sequence databases and mass spectrometry repositories will be explored. Expanding the search across multiple data sources will increase the likelihood of resolving ambiguous residues. Specifically, Aim 2 will evaluate how sequence ambiguity and incomplete annotations affect peptide function prediction. Machine learning models will be trained on both the original UniProt dataset and the curated dataset produced in Aim 1. Comparing these models will allow us to quantify how improved data quality affects prediction accuracy and confidence.

Summary of Specific Aim 1: In Aim 1, we will create a curated peptide database by resolving unknown amino acids (X residues) and expanding incomplete functional annotations in UniProt peptide sequences. We will use external peptide databases, literature verification, and UniRef cluster consensus to identify the correct amino acids and confirm peptide functions. Multiple biological activities will be recorded for each peptide with supporting references to improve data reliability.

Specific Aim 2: Quantify and model the impact of missing and ambiguous peptide information on function prediction

Rationale: Artificial intelligence models depend on accurate sequence information and complete functional annotations. However, **peptide databases often contain unknown amino acids marked as X and incomplete activity labels.** These issues reduce prediction accuracy and reduce model confidence. The effect of these problems has not been measured carefully. In Specific Aim 1, we will create a structured peptide database with resolved amino acids and expanded functional annotations. **In Specific Aim 2, we will measure how ambiguity affects prediction performance and develop computational strategies that improve prediction reliability when information is incomplete.** Our hypothesis is that resolving ambiguous amino acids and expanding annotations will improve prediction accuracy, calibration, and stability of peptide function prediction models.

Overall Study Design and Workflow: In this aim, we will construct peptide datasets with different levels of sequence ambiguity and annotation completeness. We will train machine learning models on these datasets and compare prediction performance across conditions. We will also test uncertainty-aware prediction methods to improve reliability when sequence information is incomplete.

A schematic overview of the workflow is shown in Figure 4. The diagram illustrates how peptide sequences are collected, processed into datasets with controlled ambiguity levels, encoded into machine learning features, and evaluated using multiple prediction models and uncertainty analysis methods. This workflow allows us to directly measure how sequence ambiguity affects prediction accuracy and confidence.

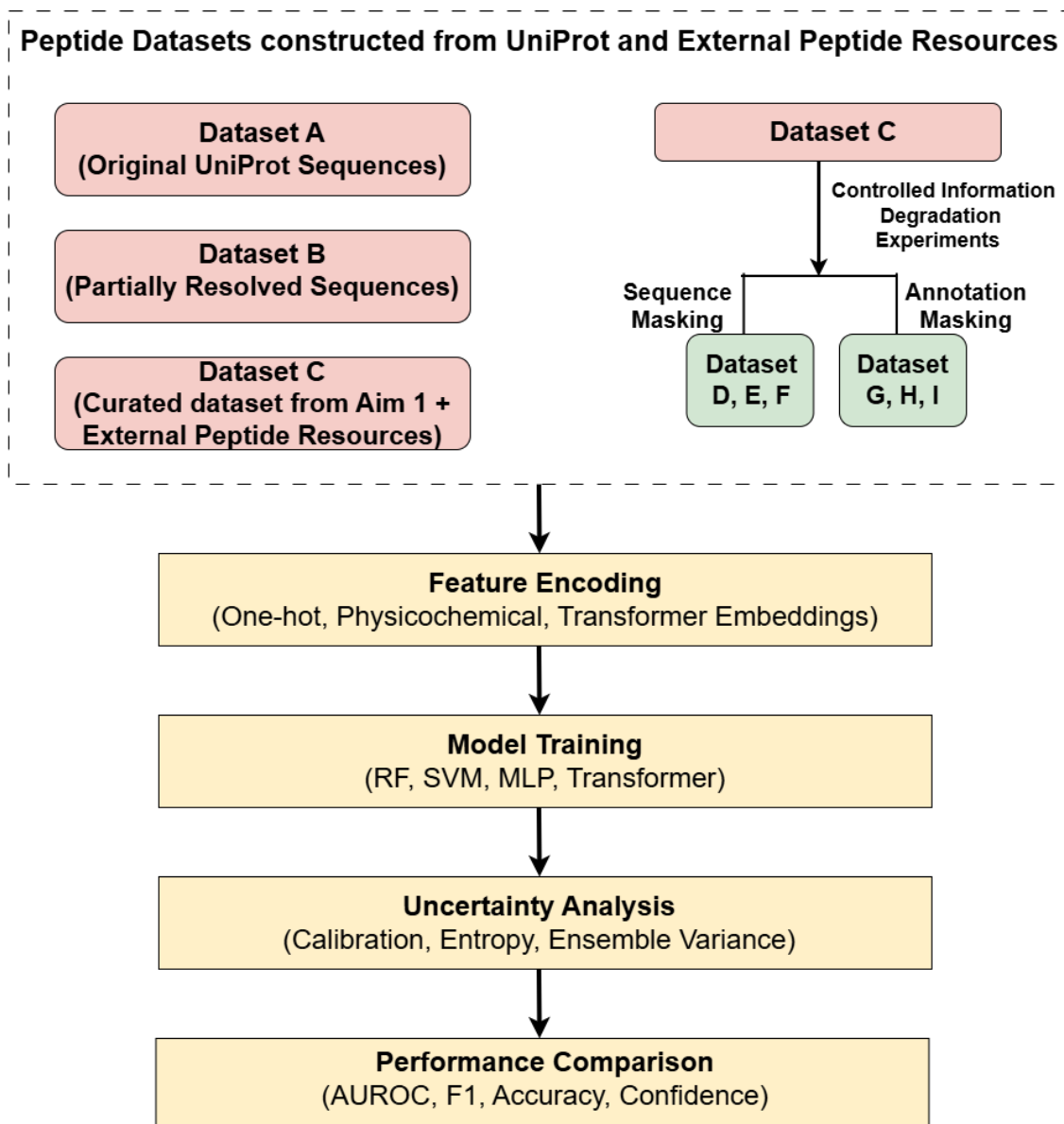


Figure 4: **Experimental workflow for evaluating the impact of ambiguous amino acids and incomplete annotations on peptide function prediction.**

The workflow shows dataset preparation from UniProt and external peptide resources, construction of datasets with controlled ambiguity levels, feature encoding, machine learning training, uncertainty estimation, and performance comparison across datasets.

2.1 Dataset Construction and Ambiguity Modeling. We will construct several peptide datasets that represent different levels of sequence completeness and annotation quality. These datasets will be derived from UniProt peptide entries and from external peptide databases such as DBAASP and UniRef clusters. Dataset A will contain original peptide sequences from UniProt with unknown amino acids and incomplete annotations. Dataset B will contain partially resolved sequences in which some ambiguous amino acids are replaced using literature-supported evidence. Dataset C will contain

curated peptide sequences generated using the resolution workflow developed in Aim 1 and supplemented with additional peptide sequences from external validated resources to ensure that modeling experiments remain feasible independent of Aim 1 outcomes.

In addition to datasets derived from UniProt and curated peptide resources, we will generate simulated ambiguity datasets to measure how increasing levels of missing sequence information affect prediction performance. In these datasets, five percent, ten percent, and twenty percent of amino acids will be randomly replaced with X to create controlled sequence ambiguity conditions. These datasets correspond to Dataset D, Dataset E, and Dataset F described in Table 1. Similarly, we will simulate incomplete annotation conditions by masking ten percent, twenty-five percent, and fifty percent of functional labels. These datasets correspond to Dataset G, Dataset H, and Dataset I. These controlled datasets allow us to systematically evaluate how missing sequence information and missing annotations influence prediction accuracy and model confidence. Datasets D through I are generated by introducing controlled levels of sequence ambiguity and annotation masking into Dataset C, which serves as the high-quality reference dataset. These simulated datasets allow systematic evaluation of how prediction performance changes as information completeness decreases.

The dataset conditions used in this aim are summarized in Table 1, which defines the sequence status, annotation completeness level, and experimental purpose for each dataset configuration used in performance evaluation.

Table 2: Dataset configurations used to evaluate the impact of sequence ambiguity and annotation completeness on prediction performance

Dataset Name	Sequence Status	Annotation Status	Experimental Role
Dataset A	Original UniProt sequences	Incomplete annotations	Baseline comparison
Dataset B	Partially resolved sequences	Expanded annotations where available	Intermediate ambiguity level
Dataset C	Curated sequences from Aim 1 and external peptide resources	Multi-label structured annotations	Primary evaluation dataset
Dataset D	5% simulated X substitutions	Complete annotations	Sensitivity analysis
Dataset E	10% simulated X substitutions	Complete annotations	Moderate ambiguity test
Dataset F	20% simulated X substitutions	Complete annotations	High ambiguity test
Dataset G	Resolved sequences	10% masked annotations	Annotation loss simulation
Dataset H	Resolved sequences	25% masked annotations	Moderate annotation loss test
Dataset I	Resolved sequences	50% masked annotations	Severe annotation loss test

2.2 Feature Extraction and Sequence Encoding. Each peptide sequence will be converted into numerical features before model training. We will use one-hot encoding to represent amino acid identity at each sequence position. We will also compute physicochemical descriptors such as hydrophobicity, charge, polarity, and molecular weight. In addition, we will extract contextual embeddings using

pretrained protein language models. These complementary feature representations will allow us to evaluate how sequence ambiguity affects both simple and advanced encoding strategies.

Ambiguous amino acids marked as X will be handled using multiple encoding strategies. In some experiments, these positions will be ignored. In other experiments, they will be replaced with averaged amino acid representations. In additional experiments, they will be represented using learned embeddings that capture uncertainty information. This comparison will help determine which encoding strategy produces the most reliable predictions under ambiguous conditions.

2.3 Machine learning Model Development. We will train several machine learning models for peptide function prediction. Classical machine learning models will include Random Forest, Support Vector Machine, and Gradient Boosting classifiers. Deep learning models will include multilayer perceptron networks, convolutional neural networks for sequence data, and transformer-based classifiers using pretrained embeddings. These models will be implemented using Python libraries including PyTorch and scikit-learn.

All models will be trained using identical preprocessing pipelines and identical dataset splits to ensure fair comparison across datasets with different ambiguity levels.

2.4 Training Procedure and Sample Size Justification. We expect to analyze approximately three thousand to five thousand peptide sequences collected from UniProt and external peptide databases. **Approximately 1000-3000 sequences** will include resolved amino acids generated using the workflow developed in Aim 1 and complementary external resources.

Each dataset will be divided into training, validation, and testing subsets using stratified sampling. **70% of the data will be used for training, 15% for validation, and 15% for testing. We will also perform 5 folds cross-validation using fixed random seeds to ensure reproducibility.**

2.5 Uncertainty-aware Prediction Framework. Prediction confidence is important when sequence information is incomplete. To address this issue, we will implement uncertainty-aware prediction methods that estimate confidence during model inference. Monte Carlo dropout sampling will be used to estimate prediction variance across multiple forward passes. Ensemble neural networks will also be trained using independent model initializations. Temperature scaling will be applied to improve calibration of prediction probabilities.

These approaches will allow us to determine whether curated peptide sequences produce more reliable predictions compared with ambiguous sequences.

2.6 Statistical Analysis and Evaluation Metrics. Model performance will be evaluated using several complementary metrics including **AUROC (Area Under the Receiver Operating Characteristic curve), Accuracy, Precision, Recall, and F1 score**. Prediction confidence will be evaluated using calibration error and entropy-based uncertainty measurements.

Statistical comparisons between datasets will be performed using paired t-tests and bootstrap confidence interval estimation with one thousand resampling iterations. These analyses will determine whether improvements in prediction performance after sequence resolution are statistically significant.

Controls and Reproducibility Strategy: To ensure valid comparisons, all experiments will use identical training pipelines and identical dataset splits across ambiguity conditions. The original UniProt dataset will serve as a baseline reference condition. Simulated ambiguity datasets will serve as controlled experimental benchmarks for evaluating the effect of missing amino acids and missing annotations. Random seed initialization will remain fixed across all experiments to ensure reproducibility of model training results.

Expected outcomes: If our hypothesis is correct, models trained on curated peptide sequences will show higher AUROC values, improved calibration reliability, and lower prediction uncertainty compared with models trained on ambiguous datasets. These improvements will confirm that resolving unknown amino acids and expanding annotations increases prediction reliability.

If prediction performance does not improve significantly after ambiguity resolution, this outcome would suggest that additional biological features such as structural descriptors or evolutionary conservation information may be required for reliable peptide function prediction. This result would still provide important insight into the limitations of sequence-only prediction models.

Summary of Aim 2: In Specific Aim 2, we will study how missing amino acids and incomplete functional annotations affect peptide function prediction using machine learning models. We will create datasets with different levels of sequence ambiguity and annotation completeness and compare prediction performance across these conditions. We will also apply uncertainty-aware prediction methods to measure how confidence changes when information is incomplete. This aim will show how improving sequence quality and annotation completeness increases the accuracy and reliability of AI-based peptide function prediction.

Overall Summary and Conclusions

This project proposes a computational framework to improve peptide sequence quality and functional annotation reliability for artificial intelligence applications. The study focuses on resolving unknown amino acids, expanding multi-functional peptide annotations, and evaluating how missing information affects machine learning prediction performance. The project also introduces uncertainty-aware modeling methods to improve prediction confidence when sequence information is incomplete. The expected outcome is a structured and extensible peptide database that supports reliable peptide function prediction and reduces ambiguity in existing peptide resources. This work may improve future AI-driven peptide discovery and support biomedical research involving antimicrobial and therapeutic peptides.